

01 The Problem & The Need

≈ 91,000 crashes
≈ 800 deaths / year

U.S. drowsy driving — NHTSA

They flag symptoms, not driver state over time. **AuraDrive reasons over time & context — not another threshold alarm.**

WHY EXISTING SYSTEMS FALL SHORT

- Camera DMS judge fixed thresholds
- false alarms on blinks, yawns & talk
- drivers distrust & switch them off

02 Requirements & Metrics

KEY REQUIREMENTS

- ≥ 30 FPS perception
- Suspicion → alert < 2 s
- 100% on device, no internet
- 12 h continuous operation
- LLM timeout → safe fallback

Targets from the requirements spec — measured on Mac , to be re profiled on Jetson (CUDA).

DID WE MEET THEM?

- ✓ EAR validated 0.20 / 0.23
- ✓ Emergency bypass: immediate
- ✓ Agent output schema valid
- ✓ Perception: 30 FPS
- ✓ Alert latency: < 2 s

03 System Pipeline — Camera Frame → Driver Alert, end to end

what we implemented · every stage: input → process → output

PERCEPTION FRONT END — continuous vision, frame by frame

0 Camera Frame Acquisition

INPUT

Camera stream

PROCESS

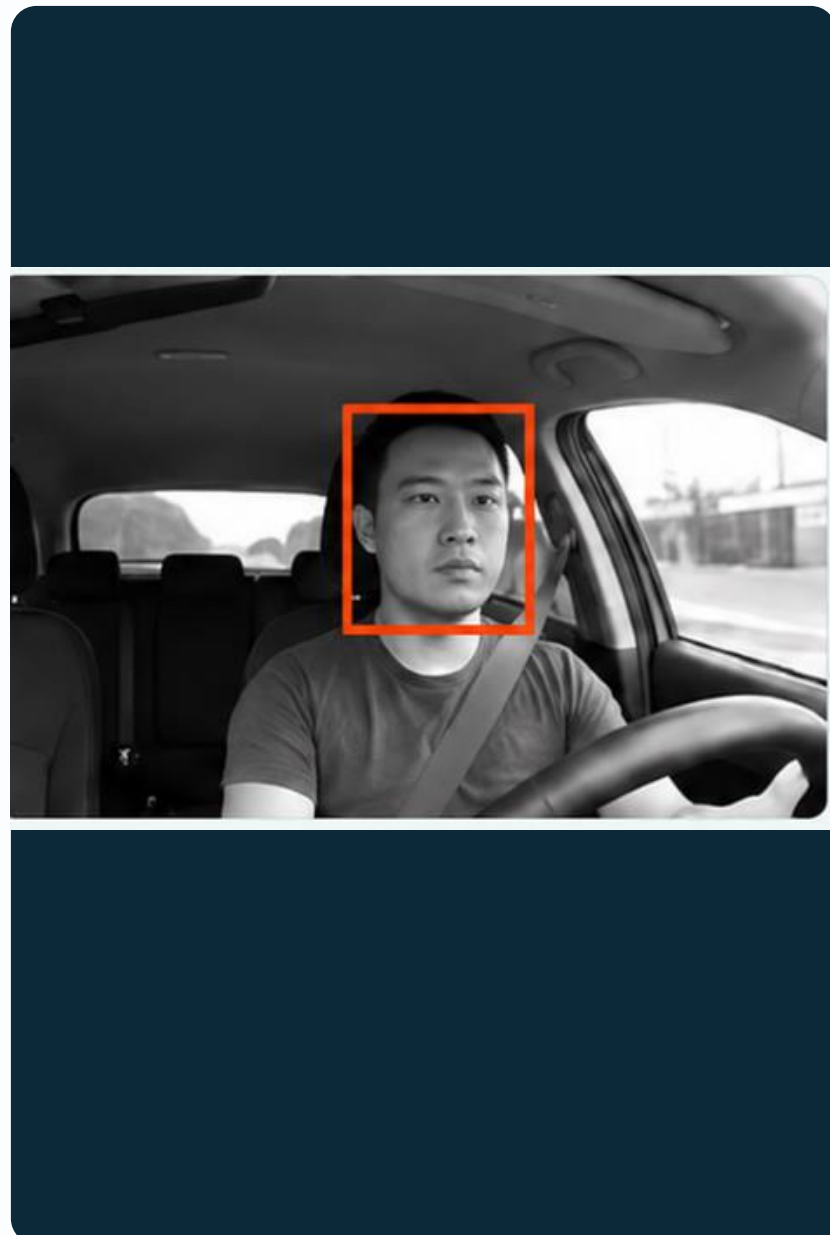
Capture frames at ≥ 30 FPS; attach a timestamp and package each frame with a unique **frame_id** — the key the whole pipeline rendezvous on.

OUTPUT

Raw RGB frame · HxWx3
+ frame_id, timestamp

OpenCV Python

1 Face Detection & Alignment



PROCESS

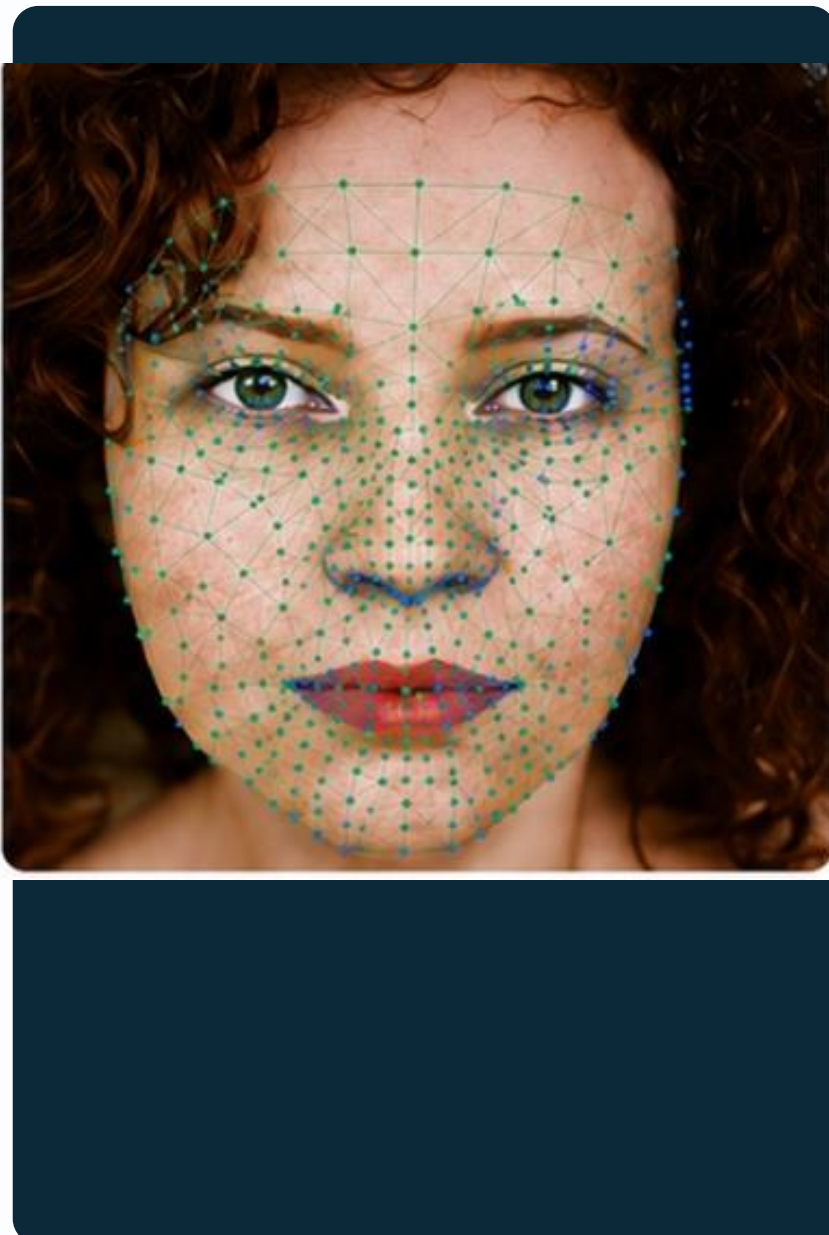
Lightweight single shot CNN (~200KB) — **detect once, then track** (5.7 ms per detection); re detect only when tracking confidence drops or the face is lost.

OUTPUT

Face BBox · aligned ROI · confidence

Lightweight CNN BlazeFace

2 3D Facial Landmark Tracking



PROCESS

Extracts **468 3D landmarks** (≈2.3 mm precision) from the face ROI; temporal continuity predicts landmarks in a **small search region** instead of full re detection.

OUTPUT

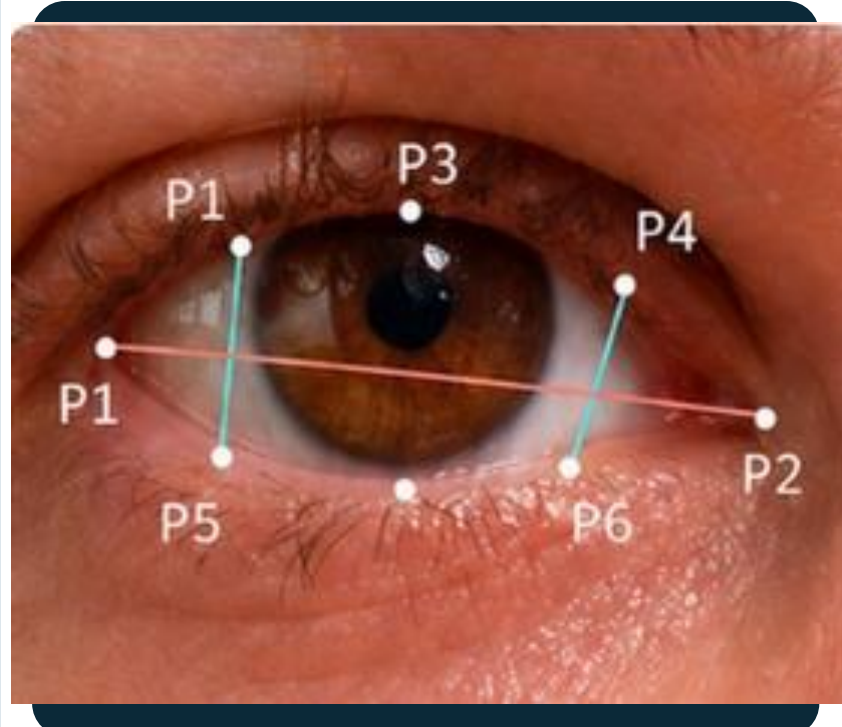
468 landmark array · (x, y, z)

MediaPipe Face Mesh

3 Per Frame Feature Extraction

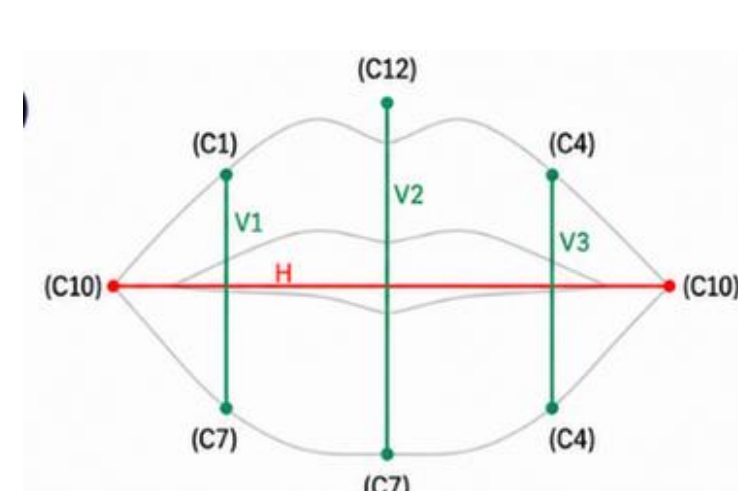
EAR — eye openness

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \cdot \|p_1 - p_4\|}$$



MAR — mouth opening

$$MAR = \frac{V_1 + V_2 + V_3}{2 \times H}$$



Head pose (pitch / roll)

$$Pitch = \arctan \frac{T_{23}}{T_{33}} \quad Roll = \arctan \frac{T_{21}}{T_{11}}$$

Euler angles solved via PnP on the 3D landmarks, measured against a **personal calibrated baseline**.

Pitch ≥ 20° Roll ≥ 15.4°

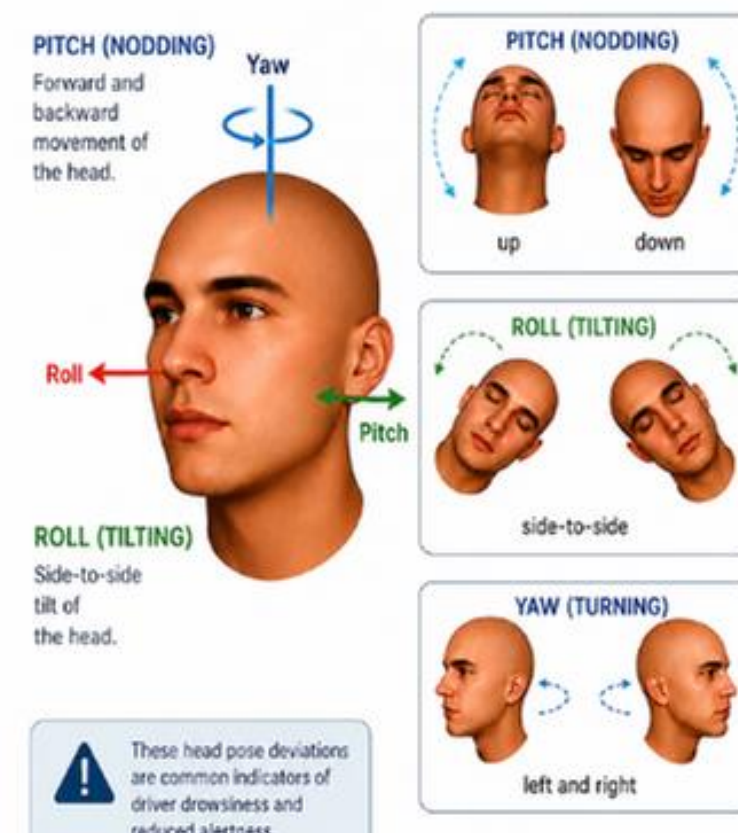
persist ≥ 0.5 s

OUTPUT

Metric vector {EAR, MAR, pitch, roll} · per frame

EAR · MAR

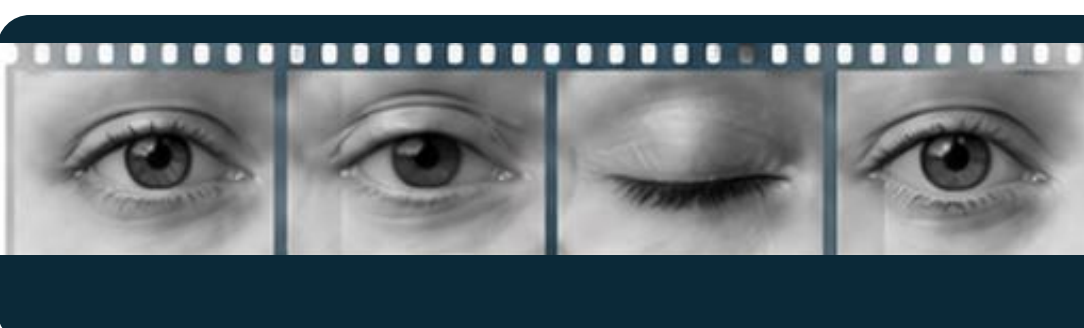
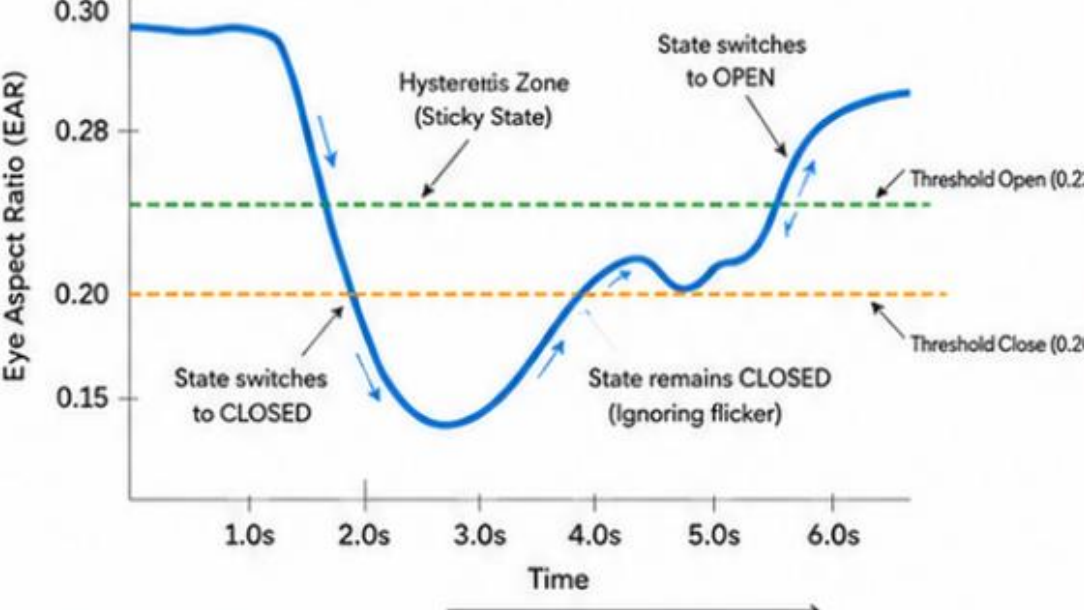
solvePnP



4 Temporal Aggregation & Events

Signal Stability: Solving the Oscillation Problem

Implementing Hysteresis via Dual Thresholds



PROCESS

Three parallel windows — **blink** (60–350 ms) / **prolonged closure** (≥ 2 s reflex) / **PERCLOS trend** (rolling 60 s) — with **hysteresis** (close 0.20 / open 0.23) to stop OPEN/CLOSED flicker. Fatigue is a process, not a single frame.

eyes closed ≥ 2000 ms? → EMERGENCY reflex

OUTPUT

Fatigue features {PERCLOS%, closure, blink, yawn, trend}

Rolling windows

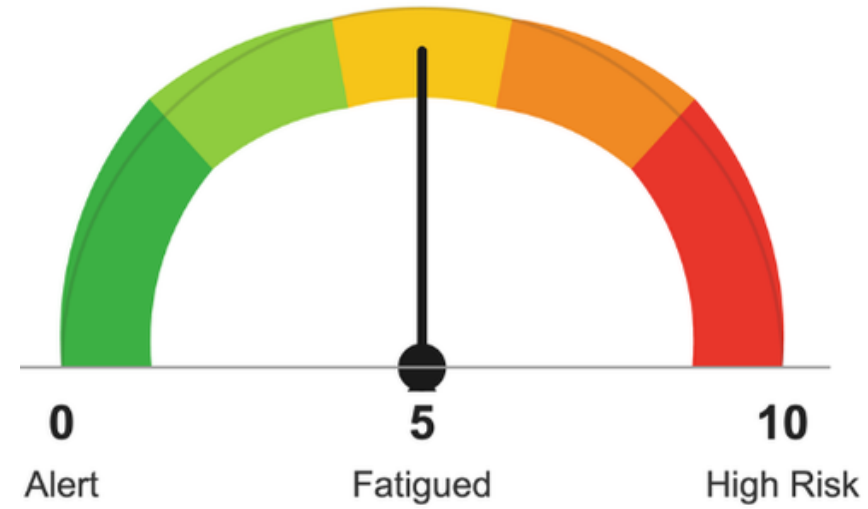
Hysteresis

▼ Aggregated Fatigue Features + Trip Context ▼

REASONING & DECISION — deterministic baseline · async agent · safety arbitrated

5 Cold Reference Engine

ALWAYS ON DETERMINISTIC SCORING



PROCESS · WEIGHTED SCORE

PERCLOS 0.51
Eye closure 0.27
Yawn 0.12
Blink 0.10

× time of day joint cap ≤ 2.5 × trip duration

Deterministic & low latency — publishes a safety baseline **immediately**, without waiting for the LLM.

OUTPUT

Cold Score 0–10 · Risk {NO_ACTION ... EMERGENCY}

Deterministic NumPy

6 Local Agent Reasoning

ASYNCHRONOUS · NON BLOCKING

Offline On Device Non Blocking

INPUT

Cold command + 10 min evidence text (System Log · Facts · Frame Narrative)

PROCESS

A local **Ollama llama3.2** model reasons about the fatigue trajectory and trip context, and returns a structured, **schema locked JSON verdict**.

Not woken at all when cold = NO_ACTION — benign frames cost zero inference.

OUTPUT · AGENT VERDICT

{Confirm · Escalate ≤ 1 tier} or Timeout

Ollama llama3.2 local HTTP

7 Safety Arbitration

→ CANONICAL DECISION

INPUT

Cold risk level + agent verdict same **frame_id** rendezvous

Cold floor cannot be weakened — a lower agent verdict is overridden.

▲ Escalation is allowed immediately — capped at one tier over cold.

▼ De escalation needs sustained evidence — TemporalGuard: fall ≤ one tier per step, min 8 s hold.

⌚ Timeout / error → cold fallback — the published baseline simply stands.

OUTPUT · CANONICAL DECISION

{NO_ACTION · GENTLE · MODERATE · URGENT · EMERGENCY}

Override rules

Temporal guard Cooldown

8 Driver Feedback & Logging

RESPONSE + AUDIT TRAIL

reflex events land here directly — **IMMEDIATE EMERGENCY, latched**

INPUT

Canonical safety decision

🔔 Alert / voice

📊 Fatigue trend

📄 Session report

📁 Event log

PROCESS

Render the matching response and audit every decision with its source (reflex / cold / agent / override).

OUTPUT

Alert Event + Session Record

Alert UI

Data logging

Analytics

Cold Score + Risk Level

ALWAYS ON · DETERMINISTIC · < 1 ms PUBLISH

published by stage 5 the moment it is computed — consumed by stage 7 as the hard safety floor

04 Testing & Challenges

TEST SCENARIOS

Glasses Yawn vs. talking Night / IR light Occlusion Latency Offline sim

CHALLENGES → SOLUTIONS

- **LLM latency 5–15 s** → cold baseline acts now; the arbiter holds the agent verdict until a new one arrives
- **Threshold false alarms** → escalate only safety floor + LLM context reasoning
- **Yawning vs. speech ambiguity** → eye narrowing co signal + MAR variance separates them
- **Dev/target hardware gap** → Metal on Mac now, CUDA on the Jetson target
- **First runs froze — alerts ~1 min late** → the camera loop was blocked while the agent was thinking; made it async
- **Agent reasoning takes time by nature** → always answer now: the cold decision is the worst case fallback

06 Conclusions & Future Work

A context aware, agentic fatigue monitor running in real time, fully on device, with **layered safety**. The combination is the architecture **a testing and safety scaffold wrapped around an inherently unreliable reasoning component**: deterministic layers set the floor, the LLM adds contextual intelligence one bounded tier at a time.

SYSTEM CONTRIBUTION

AuraDrive is a **hybrid edge AI fatigue detection system** that turns facial signals into **temporal fatigue evidence**, combines deterministic scoring with contextual local agent reasoning, and enforces a **safety controlled final decision**.

hybrid reasoning safety constrained

temporal context

All inference runs locally — no cloud, and raw video is never stored or transmitted.

05 Technologies & Tools

NO — GENTLE — MODERATE — URGENT — EMERGENCY

Python

runtime & orchestration

OpenCV

camera & vision I/O

MediaPipe

468 landmark face mesh

asyncio

one thread concurrency

NumPy

deterministic scoring

Llama 3.2

local reasoning · Ollama

Next Steps

- ✓ **NVIDIA Jetson** deployment + power profiling
- ✓ **User study** intervention effectiveness by cognitive state